

Does this pull request do what it says?

Every falsifiable claim in a PR, verified by **its own IBM Bob subagent**, in parallel — with the line of code that settles it.

THE PROBLEM

The half of code review nobody budgets for

Reading the diff is the visible half. The other half is checking that **every claim the author made is true in the code**. Under load, reviewers skip it. Here is a senior maintainer doing it by hand, this month:

KeeperHub PR #2476 · final review

“I compared each file at b957594 against 62edf7b... Does it match the description: Matches. Every behavioural claim in the body holds.”

WHAT IT DOES

Turn prose into claims, then check every one at once

01

Extract

Read the description and the review thread. Keep only claims a reader could falsify; drop intent, opinion and future tense.

02

Fan out

One Bob subagent per claim, running in parallel, each with full repository context and none seeing the others.

03

Verify

Each returns holds, does not hold or cannot tell — and the file:line that settles it. Cannot tell is never dressed up as holds.

04

Report

A ledger ordered by what should stop a merge: failures first, unchecked next, passes folded away.

Each Bob capability carries a different part of the work

Document understanding	The claims live in prose — a PR body and a review thread — not in a list.
Full repository context	The evidence for one claim is spread across files and survives a rebase.
Subagents	One per claim, isolated: a hard claim stalls only itself, and no verdict biases another.
Parallel tasks	N independent searches at once — the difference between once-per-PR and every push.

Built inside Bob IDE across five sessions. The prompts are in the repository in the order they ran; nothing in src/ was written by hand.

THE RESULT ON PR #2476

Twenty-nine claims. One of them was false.

29

claims extracted

28

hold, with file:line

1

does not hold

0

left unchecked

the sentence that failed · PR description

“...fetched through safeFetch behind an **always-on** assertUrlIsPublic
SSRF check...”

THE FINDING

The Test button skipped the guard the description promised

× DOES NOT HOLD

Every outbound request passes through `assertUrlIsPublic` as an always-on SSRF check.

EVIDENCE `plugins/predge/test.ts:31`

IMPACT The connection test fetched any URL the operator typed and reported its HTTP status — a probe of private hosts.

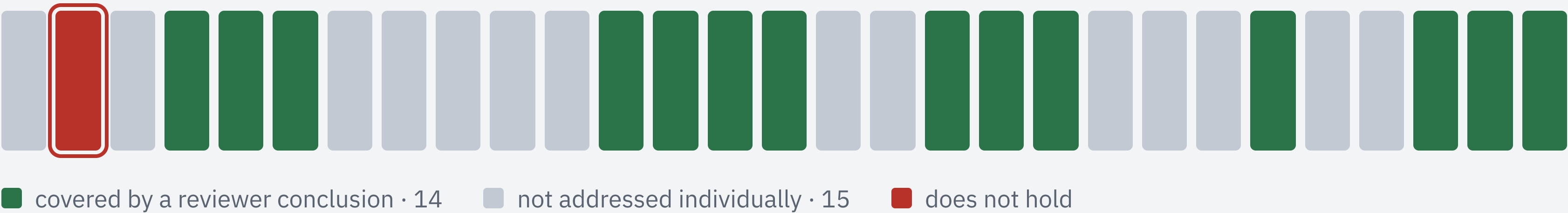
STATUS Fixed with six tests, and disclosed in the PR thread by us, the same night.

```
1025 const rawUrl =
1026   credentials.PREDGE_SIGNAL_URL?.trim() || DEFAULT_...
1027 const baseUrl = stripTrailingSlashes(rawUrl);
1028
1029 // Read-only: the published keyset confirms the ...
1030 // and is serving Predge signing keys.
1031 const response = await fetch(`${baseUrl}/.well-known/...
1032   method: "GET",
```

No `assertUrlIsPublic`. No `safeFetch`. We wrote this code.

The human pass was thorough — and partial

The final review names **four conclusions**. They cover **14 of the 29 claims**, and the tool agrees on all 14. The other 15 were never addressed one by one — and the failing claim is among them.



WHERE IT BREAKS

We tried to break our own tool. It broke twice.

FALSE HOLDS

A cheaper verifier missed the same bug

Run on the diff alone to save budget, a subagent found the guarded path, stopped, and said **holds**.

“Every request is guarded” is universal: one example cannot confirm it, only a counterexample refutes it. Universal claims now must search every call site and list what they checked.

live c02 does not hold · test.ts:31

checked 4 locations · 15.1 s

OUT OF REACH

It only checks what someone claimed

The worst defect we know of here — two canonicalisation functions that disagree on undefined keys — appears in no sentence of the PR.

No claim, so no subagent looked. A boundary of the method, not a bug in Bob — and the same reason four review rounds walked past it.

BUSINESS VALUE

Cheap enough to run on every push

10–15 s

per claim, measured on live Bob Shell
runs

0.07–0.19

Bobcoins per claim, measured

15 of 29

claims the human review never checked
individually

A reviewer's attention goes to the one claim that fails and the ones nobody could check — not to re-reading twenty-eight that hold.

Measured on five live runs, including the failing claim. The full 29-claim live batch did not complete: the 40-Bobcoin allowance ran out, mostly on one long IDE session. The demo replays the verdicts Bob's subagents produced.

Every claim, checked — and the ones nobody can check, named.

LIVE DEMO

[CLAIM-CHECK-FIVE.VERCEL.APP](https://claim-check-five.vercel.app)

CODE

[GITHUB.COM/PREDGEAI/CLAIM-CHECK](https://github.com/predgeai/claim-check)

THE PR

[GITHUB.COM/KEEPERHUB/KEEPERHUB/PULL/2476](https://github.com/keeperhub/keeperhub/pull/2476)

THE FIX

[PREDGE-AI/KEEPERHUB · 903C661](https://github.com/predge-ai/keeperhub/pull/903C661)